

Moch Rakha Aryaputra

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EDUCATION

University of California, San Diego

San Diego, CA

Bachelor of Science in Mathematics/Applied Science

Expected Graduation: June 2029

Minor in Data Science

Cumulative GPA: 4.0 / 4.0

- Fully funded by Indonesia's Ministry of Higher Education and Technology
- Provost Honors, Triton Scholars Award recipient, \$40,000 merit-based funding

HONORS AND AWARDS

- Gold Medal — Indonesian National Mathematical Olympiad (2021)
- Bronze Medal — Indonesian National Mathematical Olympiad (2024)
- Winner — Academy of Champions (National TV competition, 1M+ viewers)

PROJECTS

Sporisk

- Built a full-stack safety web application that aggregates real-time weather data (Open-Meteo) and air quality data (EPA AQS) to generate ML-based health risk scores for outdoor activities
- Engineered a geospatial risk pipeline using OpenStreetMap and California GeoJSON to surface location-specific conditions and generate natural language safety recommendations via Gemini API
- Submitted as part of HackMerced at UC Merced under the CITRIS track, health for social good.

TREZO: Automatic Trash Bin Based on Sensors

- Designed an Arduino-based prototype using sensors, Programmed microcontroller logic for automated lid response
- Contributed to school environmental management, Learned Arduino integration.

LEADERSHIP

STEM Discovery

Semarang, Indonesia

President

Sep 2024

- Led and coordinated member and volunteer activities
- Delivered live demonstrations of Tesla Coil experiments to junior high school students
- Increased STEM participation among students

Youth Mastery

Semarang, Indonesia

Team Leader

Jun 2024 – Jan 2025

- Led a team of 5 members and 4 volunteers, working on weekly videos of our actions
- Spent 3 hours per week and educated hundreds students from several schools
- Helped students develop productivity and learning strategies

Mathatala

Semarang, Indonesia

Founder

Aug 2023 – Jun 2025

- Initiated and handled an organization to improve high school students' math skill
- Created and gave 1 math material to teach students for 4 hours every week

WORK EXPERIENCE

KTOM

Remote, Indonesia

Question Division

May 2024 – May 2025

- Authored original high-difficulty math competition problems
- Contributed to national-level olympiad preparation materials
- Assisted in development and distribution of olympiad books

ADDITIONAL INFORMATION

Language : English, Indonesian

Technical : Python, Visual Code Studio, Git/Github, NumPy

Interests : Basketball, F1, Video Game